

A Proof of Conjecture 000000000034

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Abstract

Every finite colouring of $\mathbb{N}$ contains a monochromatic three-term geometric progression x, xr, xr^2 with integer ratio $r > 1$. The proof is the standard reduction to van der Waerden's theorem along the powers of two. The argument is formalized in Lean 4 against Mathlib, and the formalization is unconditional: van der Waerden's theorem is derived from Mathlib's Hales–Jewett theorem rather than assumed.

1 Statement

Theorem 1. *Let $c : \mathbb{N} \rightarrow K$ be any colouring of the natural numbers by a finite set K of colours. Then there exist a positive integer x and an integer $r > 1$ with*

$$c(x) = c(xr) = c(xr^2).$$

Since the witness produced below satisfies $x \geq 1$, it is immaterial whether $\mathbb{N}$ is taken to contain 0.

2 Proof

We use van der Waerden's theorem in the following form.

Lemma 2 (van der Waerden, three-term case). *For every colouring $c' : \mathbb{N} \rightarrow K$ by a finite set of colours there exist $a \geq 0$ and $d > 0$ with*

$$c'(a) = c'(a + d) = c'(a + 2d).$$

Proof of Theorem 1. Define a colouring of the exponents by

$$c'(m) = c(2^m), \quad m \in \mathbb{N}.$$

This is again a colouring by the finite set K , so Lemma 2 gives $a \geq 0$ and $d > 0$ with

$$c(2^a) = c(2^{a+d}) = c(2^{a+2d}).$$

Put

$$x = 2^a, \quad r = 2^d.$$

Then $x \geq 1$, and $r = 2^d > 1$ because $d > 0$. The exponent laws give

$$2^{a+d} = 2^a 2^d = xr, \quad 2^{a+2d} = 2^a (2^d)^2 = xr^2,$$

so the displayed equalities read $c(x) = c(xr) = c(xr^2)$. As r is an integer greater than 1, this is a monochromatic geometric progression of the required form. $\square$

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Remark 3. Nothing is special about the base 2: any integer base $b \geq 2$ works, and applying van der Waerden’s theorem to k -term arithmetic progressions instead yields monochromatic geometric progressions $x, xr, \dots, xr^{k-1}$ of every length k . Only the three-term case is formalized below.

3 Formal verification

The accompanying Lean 4 project formalizes Theorem 1 against Mathlib. The formalization is *unconditional*: van der Waerden’s theorem is not taken as a hypothesis.

- `vanDerWaerden3` — Lemma 2, obtained from Mathlib’s `Combinatorics.exists_mono_homothetic_copy` applied to the finite set $S = \{0, 1, 2\} \subseteq \mathbb{N}$. That theorem — a corollary of the Hales–Jewett theorem, which Mathlib proves — produces $a > 0$, b and a colour k with $c'(a \cdot s + b) = k$ for every $s \in S$, which is exactly a monochromatic three-term arithmetic progression with first term b and common difference a .
- `exists_mono_geometric_progression` — Theorem 1 for an arbitrary finite colour type:

$$\forall \kappa \text{ finite}, \forall c : \mathbb{N} \rightarrow \kappa, \quad \exists x r, 0 < x \wedge 1 < r \wedge c x = c(xr) \wedge c(xr) = c(xr^2).$$

- `exists_mono_geometric_progression_fin` — the same statement for a colouring $\mathbb{N} \rightarrow \text{Fin } q$, matching the “ q colours” phrasing of the conjecture.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms exists_mono_geometric_progression` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.